

MATHS — GEOMETRY & MEASUREMENT FOCUS*Summit Stretch Paper***Instructions to candidates**

- Write your name and date in the spaces provided below.
- Answer ALL questions in the spaces provided.
- You may NOT use a calculator for this paper.
- Where a question gives options, write the letter of your answer in the answer box.

Candidate name _____

Date _____

Time started / finished _____

1.Three angles meet at a point. Two of them are 110° and 95° . What is the size of the third angle?

[1 mark]

- A. 145°
- B. 155°
- C. 165°
- D. 150°

2.Two angles lie on a straight line. One is 62° . What is the other angle, in degrees?

[1 mark]

3.An isosceles triangle has an apex angle of 40° . What is the size of each of the two equal base angles?

[1 mark]

- A. 70°
- B. 80°
- C. 60°
- D. 140°

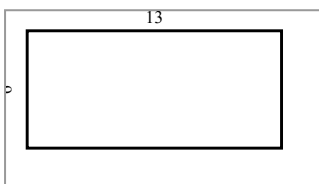
4.Two straight lines cross. One of the four angles formed is 128° . What is the size, in degrees, of an angle next to it on the same straight line?

[1 mark]

5.

A rectangular field is 13 m long and 6 m wide. What is its perimeter?

[1 mark]

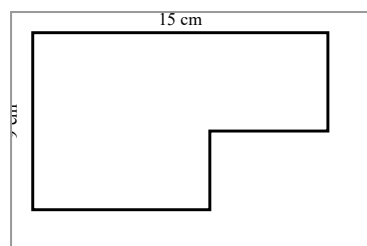


Rectangular field

6.

The L-shape has an outer width of 15 cm and outer height of 9 cm, with a 6 cm by 4 cm rectangle cut from the bottom-right corner. What is the area of the shape?

[2 marks]



L-shaped compound area

7.

The interior angles of a regular polygon add up to 900° . How many sides does the polygon have?

[1 mark]

- A. 6
- B. 7
- C. 8
- D. 9

8.

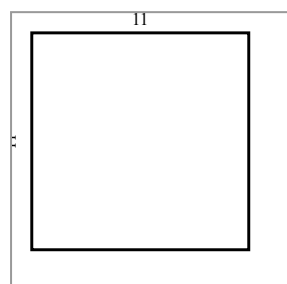
A triangle has a base of 14 cm and a height of 9 cm. What is its area, in cm squared?

[1 mark]

9.

A square has a side length of 11 cm. What is its area?

[1 mark]

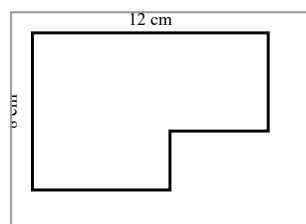


Square

10.

The L-shape has an outer width of 12 cm and outer height of 8 cm, with a 5 cm by 3 cm rectangle cut from the bottom-right corner. What is the perimeter of the shape?

[2 marks]



L-shaped compound perimeter

11.

A quadrilateral has three angles of 85° , 102° and 77° . What is the fourth angle?

[1 mark]

- A. 86°
- B. 96°
- C. 106°
- D. 116°

12.

Which of these shapes has exactly 4 lines of symmetry?

[1 mark]

- A. Square
- B. Rectangle
- C. Equilateral triangle
- D. Regular pentagon

13.

A cuboid measures 6 cm by 4 cm by 3 cm. What is its volume, in cm cubed?

[1 mark]

14.

A rectangular lawn measures 9 m by 6 m. It is surrounded on all sides by a path 1 m wide. What is the area of the path alone, in m squared? [2 marks]

15.

How many grams are there in 3.5 kg?

[1 mark]

- A. 350 g
- B. 3500 g
- C. 35 g
- D. 35000 g

16.

Convert 2400 ml into litres.

[1 mark]

17.

A stretch of road is 5.2 km long. How many metres is this?

[1 mark]

- A. 520 m
- B. 5200 m
- C. 52 m
- D. 52000 m

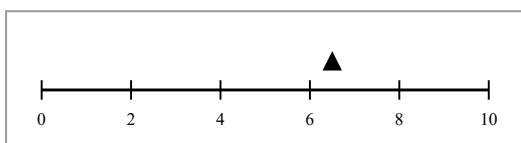
18.

A water tank holds 18 litres. The water is poured out using cups that hold 250 ml each. How many full cups can be poured? [2 marks]

19.

The arrow on the scale points to a mass, in kg. What mass does it show?

[1 mark]



Mass scale (kg)

20.

A ribbon is 4.5 m long. It is cut into pieces that are each 30 cm long. How many complete pieces can be cut?

[1 mark]

21. The table shows the mass of four parcels. Which parcel is the heaviest? [1 mark]

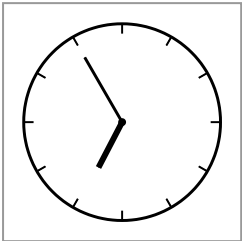
Parcel	Mass
A	1200 g
B	1.19 kg
C	1090 g
D	1.25 kg

Parcel masses

- A. Parcel A
- B. Parcel B
- C. Parcel C
- D. Parcel D

22. A bag contains 3.2 kg of flour. Each batch of bread uses 750 g of flour. After making as many complete batches as possible, how many grams of flour are left over? [2 marks]

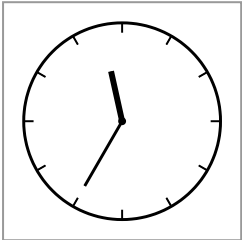
23. What time does the clock show? [1 mark]



Clock face

- A. 6:55
- B. 7:55
- C. 5:55
- D. 6:45

24. A coach departs at the time shown on the clock and arrives 47 minutes later. What time does it arrive? [1 mark]



Departure time

25. A digital clock shows 14:45. What is this in 12-hour clock time? [1 mark]

- A. 2:45 am
- B. 2:45 pm
- C. 4:45 pm
- D. 12:45 pm

26. A film starts at 19:50 and lasts 2 hours 35 minutes. What time does it finish? [2 marks]

27. How many minutes are there between 09:15 and 11:05? [1 mark]

28. Buses leave a stop every 18 minutes, starting at 08:06. What time does the 5th bus leave? [2 marks]

29. Jess pays for a £3.65 magazine with a £5 note. How much change does she receive? [1 mark]

30. The table shows stationery prices. What is the total cost of 3 pens and 2 rulers? [2 marks]

Item	Price
Pen	£0.85
Ruler	£1.20

Stationery prices

31. A jacket usually costs £40. In a sale it is reduced by 15%. What is the sale price? [1 mark]

32. Three friends share a restaurant bill of £27.60 equally, and they also add a £6 tip to be shared equally. How much does each friend pay in total? [2 marks]

33. The table shows three pack sizes of the same rice. Which pack gives the best value for money? [2 marks]

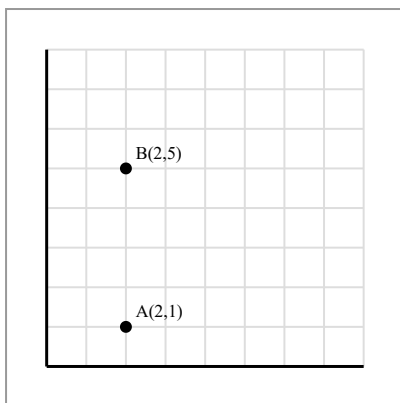
Pack size	Price
250 g	£2.00
400 g	£3.20
600 g	£4.50

Rice pack prices

- A. 250 g pack
- B. 400 g pack
- C. 600 g pack

34. Priya saves £4.50 every week. How many weeks will it take her to save £81? [1 mark]

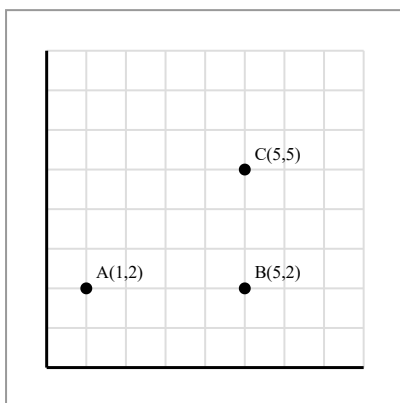
35. Points A and B are plotted on the grid. What is the distance between A and B? [1 mark]



Points A and B

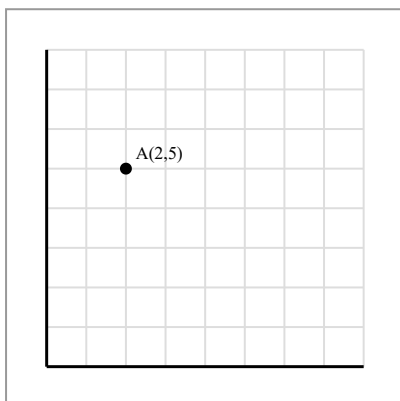
- A. 3 units
- B. 4 units
- C. 5 units
- D. 6 units

36. Three vertices of a rectangle are A(1, 2), B(5, 2) and C(5, 5), as plotted on the grid. What are the coordinates of the fourth vertex, D? [1 mark]



Rectangle ABC (D missing)

37. Point Q, plotted on the grid, is at (2, 5). It is reflected in the horizontal line $y = 3$. What are the coordinates of its image? [1 mark]

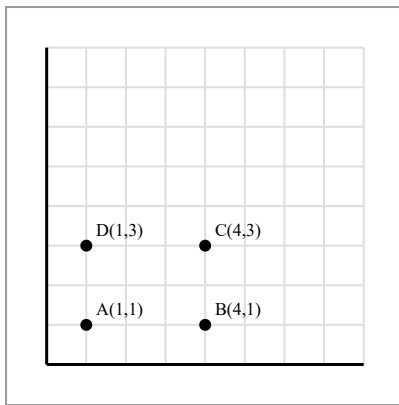


Point Q before reflection

38.

A rectangle has vertices at (1, 1), (4, 1), (4, 3) and (1, 3), as plotted on the grid. What is its area?

[2 marks]

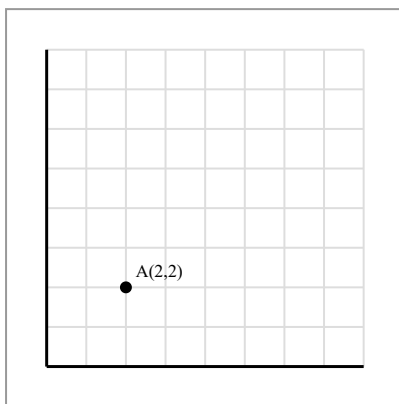


Rectangle on a grid

39.

Point P, plotted on the grid, is at (2, 2). It is translated 3 units right and 2 units up. What are the coordinates of its new position?

[1 mark]

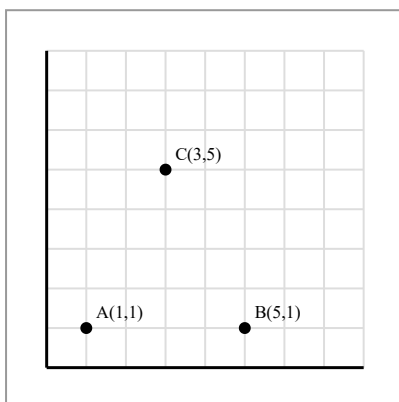


Point P before translation

40.

A triangle has vertices at A(1, 1), B(5, 1) and C(3, 5), as plotted on the grid. What is the area of the triangle?

[2 marks]

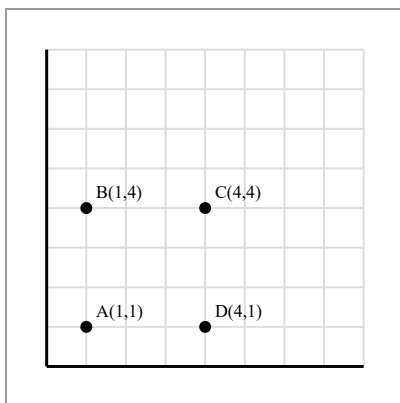


Triangle ABC

41.

A square, plotted on the grid, has vertices at $(1, 1)$, $(1, 4)$, $(4, 4)$ and $(4, 1)$. What is its area?

[1 mark]



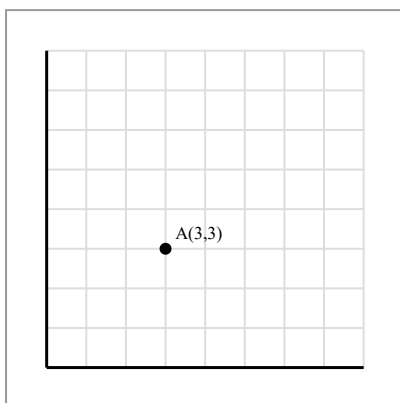
Square on a grid

- A. 6
- B. 9
- C. 12
- D. 16

42.

Point R, plotted on the grid, is at $(3, 3)$. It is translated 2 units left and 3 units up. What are the coordinates of its new position?

[1 mark]



Point R before translation

ANSWER KEY & MARK SCHEME

Q	ID	Answer	Mark scheme
1	mq1	155°	Angles at a point sum to 360°. $110 + 95 = 205$. $360 - 205 = 155^\circ$.
2	mq2	118	Angles on a straight line sum to 180°. $180 - 62 = 118^\circ$.
3	mq3	70°	Angles in a triangle sum to 180°. $180 - 40 = 140$. The two equal base angles share this: $140 / 2 = 70^\circ$ each.
4	mq4	52	Angles on a straight line sum to 180°. $180 - 128 = 52^\circ$.
5	mq5	38	Perimeter = $13 + 6 + 13 + 6 = 38$ m.
6	mq6	111	Full rectangle area = $15 \times 9 = 135$ cm squared. Cut-out area = $6 \times 4 = 24$ cm squared. Shape area = $135 - 24 = 111$ cm squared.
7	mq7	7	Interior angle sum = $(n - 2) \times 180^\circ$. $900 / 180 = 5$, so $n - 2 = 5$, giving $n = 7$.
8	mq8	63	Area of a triangle = $(\text{base} \times \text{height}) / 2 = (14 \times 9) / 2 = 126 / 2 = 63$ cm squared.
9	mq9	121	Area of a square = side $\times$ side = $11 \times 11 = 121$ cm squared.
10	mq10	40	Cutting a rectangular notch from a corner does not change the total perimeter of the bounding rectangle, because the two removed edges are replaced by two new edges of the same combined length. Perimeter = $2 \times (12 + 8) = 40$ cm.
11	mq11	96°	Angles in a quadrilateral sum to 360°. $85 + 102 + 77 = 264$. $360 - 264 = 96^\circ$.
12	mq12	Square	A square has 4 lines of symmetry (2 diagonals, 1 vertical, 1 horizontal). A rectangle has 2, an equilateral triangle has 3, and a regular pentagon has 5.
13	mq13	72	Volume of a cuboid = length $\times$ width $\times$ height = $6 \times 4 \times 3 = 72$ cm cubed.
14	mq14	34	Outer dimensions including the path = $11 \text{ m} \times 8 \text{ m} = 88$ m squared. Lawn area = $9 \times 6 = 54$ m squared. Path area = $88 - 54 = 34$ m squared.
15	mq15	3500 g	1 kg = 1000 g, so $3.5 \text{ kg} = 3.5 \times 1000 = 3500$ g.
16	mq16	2.4	1000 ml = 1 litre, so $2400 \text{ ml} = 2400 / 1000 = 2.4$ litres.
17	mq17	5200 m	1 km = 1000 m, so $5.2 \text{ km} = 5.2 \times 1000 = 5200$ m.
18	mq18	72	18 litres = 18000 ml. $18000 / 250 = 72$ full cups.
19	mq19	6.5	The arrow sits exactly halfway between the 6 kg and 8 kg marks, so it shows 6.5 kg.
20	mq20	15	$4.5 \text{ m} = 450 \text{ cm}$. $450 / 30 = 15$ complete pieces.
21	mq21	Parcel D	Convert all masses to grams: A = 1200 g, B = 1190 g, C = 1090 g, D = 1250 g. The greatest value is 1250 g, so Parcel D is heaviest.
22	mq22	200	$3.2 \text{ kg} = 3200 \text{ g}$. $3200 / 750 = 4$ remainder 200, since $4 \times 750 = 3000$. So 200 g are left over.
23	mq23	6:55	The minute hand points to 11 (55 minutes) and the hour hand sits just before 7, showing 6:55.
24	mq24	12:22	$11:35 + 25 \text{ minutes} = 12:00$. $12:00 + \text{remaining } 22 \text{ minutes} = 12:22$.
25	mq25	2:45 pm	14:45 is after 12:00 midday, so subtract 12 hours: $14:45 - 12:00 = 2:45 \text{ pm}$.
26	mq26	22:25	$19:50 + 2 \text{ hours} = 21:50$. $21:50 + 35 \text{ minutes} = 22:25$.
27	mq27	110	09:15 to 11:15 is 120 minutes. 11:05 is 10 minutes before 11:15, so $120 - 10 = 110$ minutes.
28	mq28	09:18	The 5th bus is 4 intervals after the 1st: $4 \times 18 = 72$ minutes. $08:06 + 72 \text{ minutes} = 08:06 + 1 \text{ hour } 12 \text{ minutes} = 09:18$.
29	mq29	£1.35	$\text{£}5.00 - \text{£}3.65 = \text{£}1.35$.
30	mq30	£4.95	3 pens = $3 \times \text{£}0.85 = \text{£}2.55$. 2 rulers = $2 \times \text{£}1.20 = \text{£}2.40$. Total = $\text{£}2.55 + \text{£}2.40 = \text{£}4.95$.
31	mq31	£34	15% of $\text{£}40 = \text{£}6$. Sale price = $\text{£}40 - \text{£}6 = \text{£}34$.
32	mq32	£11.20	Total to share = $\text{£}27.60 + \text{£}6.00 = \text{£}33.60$. Each friend pays $\text{£}33.60 / 3 = \text{£}11.20$.
33	mq33	600 g pack	Price per 100 g: 250 g pack = $\text{£}2.00 / 2.5 = \text{£}0.80$. 400 g pack = $\text{£}3.20 / 4 = \text{£}0.80$. 600 g pack = $\text{£}4.50 / 6 = \text{£}0.75$. The 600 g pack has the lowest price per 100 g, so it is best value.
34	mq34	18	$\text{£}81 / \text{£}4.50 = 18$ weeks.
35	mq35	4 units	A is at (2, 1) and B is at (2, 5). Since the x-coordinates match, the distance is the difference in y-values: $5 - 1 = 4$ units.
36	mq36	(1, 5)	In a rectangle, D must share its x-coordinate with A ($x = 1$) and its y-coordinate with C ($y = 5$), giving $D = (1, 5)$.
37	mq37	(2, 1)	Q is 2 units above the line $y = 3$ ($5 - 3 = 2$). The image is 2 units below the line: $3 - 2 = 1$. The x-coordinate stays the same, so the image is (2, 1).
38	mq38	6	Width = $4 - 1 = 3$ units. Height = $3 - 1 = 2$ units. Area = $3 \times 2 = 6$ square units.

39	mq39	(5, 4)	Add 3 to the x-coordinate: $2 + 3 = 5$. Add 2 to the y-coordinate: $2 + 2 = 4$. New position = (5, 4).
40	mq40	8	Base AB = $5 - 1 = 4$ units (along $y = 1$). Height = vertical distance from C to line AB = $5 - 1 = 4$ units. Area = (base x height) / 2 = $(4 \times 4) / 2 = 8$ square units.
41	mq41	9	Side length = $4 - 1 = 3$ units. Area = $3 \times 3 = 9$ square units.
42	mq42	(1, 6)	Subtract 2 from the x-coordinate: $3 - 2 = 1$. Add 3 to the y-coordinate: $3 + 3 = 6$. New position = (1, 6).